Shiv Raj Bangad

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Education

Purdue University, Daniels School of Business, Master of Science in Business Analytics and Information Management – West Lafayette, IN — Dec 2026

- **Relevant Courses:** Business Analytics, Data Mining, Machine Learning, Reinforcement Learning, Cloud Computing, Advanced Database Management, MLOps, Spreadsheet Modelling, Computational Finance
- **Certifications:** AWS Certified Cloud Practitioner, AI-900 Microsoft Azure AI Fundamentals

University of Cincinnati, Carl H. Lindner School of Business, BS in Business Analytics and BBA in Finance – Cincinnati, OH — May 2025

- **GPA:** 3.9/4.0
- **Achievements:** Lindner Business Honors: Kolodzik Business Scholar (Awarded to the top 5% of business students)

Experience

Business Intelligence Engineer Intern, **Amazon (AWS Analytics)** – Seattle, WA — May 2026 - Aug 2026

- Built an AI agent system on AWS Lambda and Bedrock AgentCore that automates end-to-end AWS feature adoption tracking, reducing delivery time from 2 days of manual engineering work to within 30 minutes.
- Engineered scalable PySpark and AWS Glue pipelines on S3 to ingest and process 10B+ row multi-terabyte datasets across 100+ service features, serving as the underlying data layer for agent execution and live dashboards.
- Designed a deterministic AST-based SQL parser & guardrail to validate 100% of LLM-generated SQL against production data contracts, systematically preventing invalid joins, double-counting, and broken schema builds.
- Deployed a containerized MCP (Model Context Protocol) tool server with SigV4 cross-account authentication, establishing a robust CI/CD pipeline to manage seamless production releases.
- Implemented an NL-to-SQL Q&A engine integrated with automated statistical anomaly detection (z-score baseline tracking) to auto-generate weekly executive deep-dice narratives.

Data Science Consultant (Capstone Project), **Google (Youtube)** – West Lafayette, IN — Jan 2026 - May 2026

- Engineered a distributed PySpark ETL pipeline to ingest, clean, and vectorize unstructured metadata across 1,000+ creator channels, reducing data processing latency by 45% and structuring high-volume channel telemetry.
- Delivered an interactive end-to-end LLM recommendation platform using RAG (Retrieval-Augmented Generation) and vector embeddings, enabling creators to query cross-channel trends in real time and driving a 15% lift in predicted engagement.
- Constructed an interactive creator analytics dashboard in Dash/Streamlit to visualize content whitespace and optimal posting cadences across 20+ video categories.
- Developed an automated benchmarking framework utilizing statistical A/B testing simulations to evaluate LLM strategy recommendations against historical performance baselines.

Financial Analyst Intern, **BDT & MSD Partners (Private Equity)** – Chicago, IL — Jun 2025 - Aug 2025

- Engineered automated cash-flow ETL pipelines using advanced SQL queries and script-driven Excel models, cutting capital workflow processing time by 30%.
- Executed automated data validation and schema checks for $100M in capital calls and distributions, ensuring 100% referential integrity across complex institutional investor databases.
- Architected normalized relational data models to consolidate fragmented portfolio metrics across $2B+ AUM, resolving structural data gaps and standardizing historical formats.
- Optimized downstream data integration, establishing a centralized SQL repository that accelerated query throughput and eliminated manual data entry for executive reporting dashboards.

Data Analyst (Cintas), **University of Cincinnati - Center for Business Analytics** – Cincinnati, OH — Jan 2025 - May 2025

- Architected an automated Pandas/Dash SKU anomaly detection dashboard, automating manual audit routines and cutting operational review time by 360+ hours annually.
- Built scalable data processing scripts incorporating statistical normalization and feature engineering to resolve skewed SKU distributions, boosting model predictive performance by 30%.
- Architected automated data validation workflows and an interactive Dash monitoring framework to continuously ingest, clean, and flag structural data discrepancies across product catalogs.

Data Analyst (HBH Holdings), **University of Cincinnati - Center for Business Analytics** – Cincinnati, OH Sep 2024 - Dec 2024

- Engineered an automated document pipeline using Python, Poppler, and Unicode NFKC normalization to extract and structure raw text from complex legal PDF and Word contracts.
- Architected an AI-powered contract Q&A assistant using Haystack and the Mistral 7B LLM, enabling non-technical stakeholders to query contract terms in plain English with sub-5 second responses.
- Implemented advanced NLP preprocessing and semantic vector indexing, increasing domain-specific search accuracy by 75% and ensuring reliable information retrieval across complex legal documents.

Data Analyst Intern, **Great American Insurance Group** – Cincinnati, OH Apr 2022 - Apr 2024

- Executed and managed monthly data pipeline processing for 70,000+ policies, managing a total premium value of approximately $1 billion.
- Designed relational Access/SQL Server databases and optimized indexing strategies, reducing analytical data extraction query times by 40%.
- Engineered automated Tableau reporting workflows, replacing legacy manual data preparation and cutting monthly analytical prep cycle time by 50%.

Projects

Data4Good National AI Competition Oct 2025 - Jan 2026

- Secured Regional Championship title by developing a custom DeBERTa model that achieved 97.5% accuracy.
- Architected an end-to-end NLP data processing framework incorporating automated tokenization, TF-IDF vectorization, and DeBERTa-v3 model fine-tuning to classify semantic relationships as factual, contradictory, or irrelevant.
- Delivered a production-ready model inference pipeline utilizing a stratified weighted ensemble to aggregate prediction logits, reducing classification error rates by 15% and ensuring robust generalization on unseen evaluation data.

Predictive Modelling for Firm Bankruptcy | Kaggle Competition Nov 2025 - Dec 2025

- Achieved 1st place out of 37 teams by building an automated ML engineering pipeline that delivered a competition-leading 0.9493 ROC-AUC score in corporate insolvency prediction.
- Engineered an "Evolutionary XGBoost" automated feature generation pipeline that systematically synthesized, mutated, and selected non-linear feature interactions across 100 generations to handle severe class imbalance.
- Constructed a scalable 3-stage Stacking Super-Learner pipeline integrating feature transformation layers, multi-model parallel inference (XGBoost, LightGBM, CatBoost), and a meta-learner execution layer.

Stock Market Return Prediction (Research Assistant) May 2024 - Aug 2024

- Built an automated financial data ingestion and processing pipeline to ingest, clean, and align macroeconomic indicators and corporate fundamental metrics for quantitative analysis.
- Developed automated feature engineering routines using Principal Component Analysis (PCA) and rolling-window normalization to eliminate multi-collinearity and feature leakage.
- Delivered a multi-model evaluation framework across BART, SVM, and Random Forest architectures, achieving a 10% boost in out-of-sample R2 prediction accuracy.

Skills

Data Engineering & Analytics: ETL/ELT Pipeline Development, AWS (Glue, EMR, Redshift, S3, Lambda, Bedrock AgentCore, QuickSight), PySpark, Apache Spark, Tableau, Power BI, MySQL, PostgreSQL, SQL Server, Azure, GCP, Hadoop

Languages & Querying: Python (Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow), SQL (PostgreSQL, MySQL, SQL Server), R, Java, Bash/Zsh

Data Platforms & Storage: Data Warehousing, Relational Databases, NoSQL, Vector Databases (ChromaDB, PGVector), S3 Parquet/Delta Storage, Distributed Storage

Machine Learning & MLOps: DeBERTa, Transformers, LLM Agent Architectures, RAG Systems, XGBoost, LightGBM, Stacking Ensembles, A/B Testing, Feature Engineering, CI/CD, Docker, Git/GitHub